arXiv:2306.03034v2 [cs.AI] 7 Jan 2024

Tackling Cooperative Incompatibility for Zero-Shot Human-AI Coordination

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Abstract

Securing coordination between AI agent and teammates (human players or AI agents) in contexts involving unfamiliar humans continues to pose a significant challenge in Zero-Shot Coordination. The issue of cooperative incompatibility becomes particularly prominent when an AI agent is unsuccessful in synchronizing with certain previously unknown partners. Traditional algorithms have aimed to collaborate with partners by optimizing fixed objectives within a population, fostering diversity in strategies and behaviors. However, these techniques may lead to learning loss and an inability to cooperate with specific strategies within the population, a phenomenon named cooperative incompatibility in learning. In order to solve cooperative incompatibility in learning and effectively address the problem in the context of ZSC, we introduce the **C**ooperative **O**pen-ended **LE**arning (**COLE**) framework, which formulates open-ended objectives in cooperative games with two players using perspectives of graph theory to evaluate and pinpoint the cooperative capacity of each strategy. We present two practical algorithms, specifically COLE_{SV} and COLE_{R}, which incorporate insights from game theory and graph theory. We also show that COLE could effectively overcome the cooperative incompatibility from theoretical and empirical analysis. Subsequently, we created an online Overcooked human-AI experiment platform, the COLE platform, which enables easy customization of questionnaires, model weights, and other aspects. Utilizing the COLE platform, we enlist 130 participants for human experiments. Our findings reveal a preference for our approach over state-of-the-art methods using a variety of subjective metrics. Moreover, objective experimental outcomes in the Overcooked game environment indicate that our method surpasses existing ones when coordinating with previously unencountered AI agents and the human proxy model. Our code and demo are publicly available at `https://sites.google.com/view/cole-2023`.

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1. Introduction

Significant advancements in artificial intelligence (AI) research have led to groundbreaking solutions such as AlphaZero (Silver, Hubert, Schrittwieser, Antonoglou, Lai, Guez, Lanctot, Sifre, Kumaran, Graepel, Lillicrap, Simonyan, & Hassabis, 2018) and Segment Everything (Kirillov, Mintun, Ravi, Mao, Rolland, Gustafson, Xiao, Whitehead, Berg, Lo, Dollár, & Girshick, 2023), showing their potential to outperform humans in various tasks. However, establishing efficient collaboration between AI and unseen partners (either human players or AI agents), a concept known as zero-shot coordination, continues to pose a significant challenge (Legg & Hutter, 2007; Hu, Lerer, Peysakhovich, & Foerster, 2020; De Peuter & Kaski, 2022). The significance of zero-shot human-AI coordination becomes evident in various real-world applications, including manufacturing (Li, Zheng, Liu, Wang, Wang, Zheng, & Wang, 2023a), autonomous vehicles (Aoki, Lin, & Rajkumar, 2021), and assistant robots (de Berardinis, Pizzuto, Lanza, Chella, Meira, & Cangelosi, 2020). The issue of *"cooperative incompatibility"* becomes particularly prominent when an AI agent is unsuccessful in synchronizing with certain previously unknown partners. As exemplified by multiplayer video games like Honor of King (Wei, Chen, Ji, Qin, Deng, Li, Wang, Zhang, Yu, Lin, Huang, Ye, Fu, & Yang, 2022) and Overcooked (Carroll, Shah, Ho, Griffiths, Seshia, Abbeel, & Dragan, 2020), the constant requirement for players to collaborate with unseen partners is evident. However, AI agents, which are trained via the maximization of rewards, tend to exhibit deterministic strategies and differ from those of human players or other AI agents, leading to a failure in coordinating with some unseen players (Ye, Chen, Zhang, Chen, Yuan, Liu, Chen, Liu, Qiu, Yu, Yin, Shi, Wang, Shi, Fu, Yang, Huang, & Liu, 2020; Gao, Liu, Wang, Lian, Wang, Li, Wang, Zeng, Wang, Wang, Fu, Yang, Huang, & Liu, 2023; Yu, Gao, Liu, Xu, Tang, Yang, Wang, & Wu, 2023).

One of the conventional methods to solve zero-shot human-AI coordination is self-play (Tesauro, 1994), which involves an iterative strategy refinement process through self-competition. Although SP can achieve equilibrium in a game (Fudenberg & Levine, 1998), strategies often develop specific behaviors and conventions to secure higher payoffs (Hu et al., 2020). As a result, a fully-converged SP strategy may face difficulties adapting to coordination with previously unencountered strategies and humans (Lerer & Peysakhovich, 2018; Hu et al., 2020; Zhao, Song, Haifeng, Gao, Wu, Sun, & Wei, 2021). To address SP's limitations, current zero-shot coordination (ZSC) approaches concentrate on enhancing strategic or behavioral diversity by incorporating population-based training (PBT) to improve adaptability (Carroll et al., 2020; Canaan, Gao, Togelius, Nealen, & Menzel, 2022; Zhao et al., 2021; Lupu, Cui, Hu, & Foerster, 2021; Charakorn, Manoonpong, & Dilokthanakul, 2023). PBT aims to boost cooperative performance with other strategies in the population, fostering zero-shot coordination with unknown strategies, which is achieved by maintaining a set of strategies to disrupt SP conventions (Tesauro, 1994) and optimizing the rewards for each pair within the population. Most state-of-the-art (SOTA) methods focus on pre-training diverse populations (Strouse, McKee, Botvinick, Hughes, & Everett, 2021; Lupu et al., 2021) or introducing handcrafted techniques (Canaan et al., 2022; Zhao et al., 2021) to excel at cooperative games by optimizing fixed objectives within the population. And a recent work LIPO (Charakorn et al., 2023) pursues solutions compatible with their partner agents but incompatible with others in the population. These approaches have effectively tackled

complex cooperative tasks, such as Overcooked (Carroll et al., 2020) and Hanabi (Bard, Foerster, Chandar, Burch, Lanctot, Song, Parisotto, Dumoulin, Moitra, Hughes, et al., 2020).

However, optimizing a fixed population-level objective, such as maximizing expected rewards within the population (Strouse et al., 2021; Lupu et al., 2021; Zhao et al., 2021) or maximizing self-reward, but minimizing the reward with other agents (Charakorn et al., 2023), does not guarantee improved coordination capabilities for strategies within the population. Specifically, although overall performance may improve, simultaneous promotion of coordination abilities within the population may not occur. This phenomenon, which we refer to as "*cooperative incompatibility in learning*", underscores the need to carefully weigh the trade-offs between overall performance and coordination ability when optimizing a fixed population-level objective.

To more effectively describe and formulate the cooperative incompatibility problem, we introduce Graphic-Form Games (GFGs) to reframe cooperative tasks from the perspective of norm-form games. In a GFG, strategies are depicted as nodes, with edge weights between nodes representing the corresponding cooperative utility of the connected strategies. Additionally, we derive Preference GFGs (P-GFGs) to profile each node's preferred counterpart in the population, where "prefer" signifies that a node achieves a higher score when cooperating with the preferred node rather than its neighbors. Using (P-)GFGs, cooperative incompatibility in a learning algorithm can be assessed based on the extent to which others prefer the updated strategy.

To address cooperative incompatibility, we propose the Cooperative Open-ended LEarning (**COLE**) framework, which iteratively generates a new strategy that approximates the best response to empirical gamescapes within P-GFGs. We have shown that COLE framework converges to the local best-preferred strategy at a Q-sublinear rate when using in-degree centrality as the preference evaluation metric. We propose two practical algorithms, COLE_{SV} and COLE_{R}, both of which comprise of a simulator, a solver, and a trainer that are incorporated to excel in two-player cooperative game Overcooked (Carroll et al., 2020). The primary distinguishing factor between COLE_{SV} and COLE_{R} is the solver component. While COLE_{SV} employs the intuitive concept of the Shapley value to evaluate the adaptability of strategies and ascertain the cooperative incompatibility distribution, COLE_{R} takes a different approach. It supplants the Shapley value with the average payoffs associated with nodes in the population as a measure of cooperative capability. The trainer aims to approximate the best responses to the cooperative incompatibility distribution mixture found in the population.

The prevailing literature of zero-shot human-AI collaboration largely concentrates on the quantitative assessment, indicated in the body of works (Carroll et al., 2020; Strouse et al., 2021; Yu et al., 2023), and often prioritizing metrics such as episode rewards and frequencies. Some methods conducted human-AI experiments to verify the zero-shot coordination performance with human players. Notwithstanding, these investigations tend to scantily scratch the surface of more profound aspects, predominantly bypassing detailed aspects of the subject like preference of human (Lou, Guo, Zhang, Wang, Huang, & Du, 2023; Strouse et al., 2021). These oversights accentuate the existing deficiencies in the contemporary experimental environments, which necessitates the urgency to introduce our proposed experimental framework. For the robust evaluation of human-AI collaboration performance, we devised an online Overcooked human-AI experimental pipeline, which